

Week 8 In-Class Worksheet

Contingency Tables and Chi-Square Tests

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Chi-Square Tests, Fisher's Exact Test, Odds Ratios

Instructions

Work with a partner to solve the following problems. Show all your work including formulas, calculations, and interpretations. You may use R to verify answers, but demonstrate understanding by showing the steps.

1 Problem 1: Chi-Square Goodness-of-Fit Test

A genetics researcher is studying flower color in pea plants. Mendelian genetics predicts a 9:3:3:1 ratio for four color types. The researcher observes the following in 160 plants:

Purple	Red	Pink	White
95	28	26	11

Part A: State the null and alternative hypotheses.

Part B: Calculate the expected counts for each color type based on the 9:3:3:1 ratio.

Total plants: $n = 160$

Expected proportions: Purple = $9/16$, Red = $3/16$, Pink = $3/16$, White = $1/16$

Color	Observed	Expected Proportion	Expected Count	$(O - E)^2/E$
Purple	95	$9/16 = 0.5625$		
Red	28	$3/16 = 0.1875$		
Pink	26	$3/16 = 0.1875$		
White	11	$1/16 = 0.0625$		

Part C: Calculate the chi-square test statistic.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} =$$

Part D: Determine the degrees of freedom and find the p-value (use R or chi-square table).

$$df = k - 1 =$$

$$\text{P-value} =$$

Part E: State your conclusion at $\alpha = 0.05$ level.

2 Problem 2: 2×2 Contingency Table - Smoking and Heart Disease

A cohort study follows 500 smokers and 500 non-smokers for 10 years to assess heart disease incidence.

	Heart Disease	No Heart Disease	Total
Smoker	75	425	500
Non-smoker	35	465	500
Total	110	890	1000

Part A: State the hypotheses for a chi-square test of independence.

Part B: Calculate the expected counts for each cell.

Formula: $E_{ij} = \frac{(\text{Row } i \text{ total}) \times (\text{Column } j \text{ total})}{\text{Grand total}}$

	Heart Disease	No Heart Disease
Smoker		
Non-smoker		

Part C: Calculate the chi-square test statistic.

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} =$$

Part D: Find the p-value and state your conclusion. ($df = (r - 1)(c - 1)$)

$df =$

P-value =

Conclusion:

Part E: Calculate the relative risk (RR) of heart disease for smokers vs non-smokers.

$$\text{Risk of heart disease among smokers} = \frac{75}{500} =$$

$$\text{Risk of heart disease among non-smokers} = \frac{35}{500} =$$

$$\text{Relative Risk} = \frac{\text{Risk (smokers)}}{\text{Risk (non-smokers)}} =$$

Interpretation:

Part F: Calculate the odds ratio (OR).

$$\text{Odds Ratio} = \frac{a \times d}{b \times c} = \frac{75 \times 465}{425 \times 35} =$$

Part G: Calculate the 95% confidence interval for the odds ratio.

$$SE_{\log(OR)} = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} =$$

$$\log(OR) \pm 1.96 \times SE_{\log(OR)} =$$

$$95\% \text{ CI for OR} = \left(e^{\text{lower}}, e^{\text{upper}} \right) =$$

Interpretation: Does the CI include 1? What does this mean?

3 Problem 3: Fisher's Exact Test

A small clinical trial tests a new treatment for a rare disease. Results:

	Improved	Not Improved	Total
Treatment	7	3	10
Control	2	8	10
Total	9	11	20

Part A: Calculate the expected counts for a chi-square test.

	Improved	Not Improved
Treatment		
Control		

Part B: Are the chi-square test assumptions satisfied? Explain.

Rule: All expected counts should be ≥ 5

Part C: Based on your answer to Part B, which test should be used?

Part D: Use R to perform Fisher's exact test and report the p-value and odds ratio.

R code:

P-value (two-sided) =

Odds Ratio =

95% CI for OR =

Part E: State your conclusion about treatment effectiveness.

4 Problem 4: Larger Contingency Table

A study examines the relationship between education level and exercise frequency among 300 adults.

	Never	Sometimes	Regularly	Total
High School	35	45	20	100
Bachelor's	25	40	35	100
Graduate	15	30	55	100
Total	75	115	110	300

Part A: What are the degrees of freedom for this chi-square test?

$$df = (r - 1) \times (c - 1) =$$

Part B: Calculate the expected count for the cell (Bachelor's, Regularly).

$$E_{\text{Bachelor's, Regularly}} = \frac{(\text{Row total}) \times (\text{Column total})}{\text{Grand total}} =$$

Part C: Using R, perform the chi-square test. Report:

R code:

χ^2 statistic =

P-value =

Part D: State your conclusion. Is there an association between education level and exercise frequency?

Part E: Looking at the data, describe the pattern you observe.

Hint: Calculate the proportion who exercise regularly in each education group.

High School: $\frac{20}{100} = 20\%$

Bachelor's:

Graduate:

Pattern:

5 Problem 5: McNemar's Test for Paired Data

A study evaluates whether a public health campaign changes people's attitudes. 60 people surveyed before and after:

Before Campaign	After Campaign		Total
	Positive	Negative	
Positive	18	12	30
Negative	22	8	30
Total	40	20	60

Part A: Why is McNemar's test appropriate here instead of a standard chi-square test?

Part B: Identify the discordant pairs.

Changed from Positive to Negative:

Changed from Negative to Positive:

Part C: Calculate McNemar's test statistic.

$$\chi^2_{\text{McNemar}} = \frac{(b - c)^2}{b + c} =$$

Part D: Find the p-value (use R or chi-square table with $df = 1$).

P-value =

Part E: State your conclusion. Did the campaign significantly change attitudes?

Part F: In which direction did attitudes change more?

Reflection Questions

1. What is the key difference between a chi-square goodness-of-fit test and a chi-square test of independence?
2. When should you use Fisher's exact test instead of a chi-square test?
3. Explain the difference between relative risk and odds ratio. When is each appropriate?
4. Why does a confidence interval for an odds ratio that includes 1.0 indicate no significant association?
5. What does it mean if a chi-square test is significant ($p < 0.05$) but the odds ratio is close to 1?

R Code Reference

Chi-square goodness-of-fit:

```
observed <- c(95, 28, 26, 11)
expected_props <- c(9/16, 3/16, 3/16, 1/16)
```

```
chisq.test(observed, p=expected_props)
```

Chi-square test of independence:

```
data <- matrix(c(75, 425, 35, 465), nrow=2, byrow=TRUE)
chisq.test(data)
```

Fisher's exact test:

```
data <- matrix(c(7, 3, 2, 8), nrow=2, byrow=TRUE)
fisher.test(data)
```

McNemar's test:

```
data <- matrix(c(18, 12, 22, 8), nrow=2, byrow=TRUE)
mcnemar.test(data)
```